A2 Level Computer Science — Semester 1 Course Overview | Victoria Park Academy

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18-Week Curriculum | Academic Year 2024 - 2025

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Qualification: A-Level Computer Science (AQA / OCR / Edexcel compatible)

Duration: 18 weeks (September – January)

Total Contact Hours: 72 hours (4 hours/week)

Prerequisite: Successful completion of AS Level Computer Science

A2 计算机科学 — 第一学期课程概述 / A2 Computer Science — Semester 1 Overview

本课程涵盖高级数据表示、通信、硬件、系统软件和人工智能。This course covers advanced data representation, communication, hardware, system software, and artificial intelligence.

Student Profile — A2 Level

By the end of Semester 1, A2 students will be able to:

Manipulate and simplify Boolean expressions using algebra and Karnaugh maps

Represent and normalise floating-point numbers, identifying sources of error

Explain the TCP/IP protocol stack and routing mechanisms in detail

Design and analyse logic circuits, adders, and sequential circuits using flip-flops

Describe memory management techniques including paging, segmentation, and virtual memory

Compare CPU scheduling algorithms and explain interrupt handling

Evaluate AI technologies including expert systems, neural networks, and machine learning

Apply ethical reasoning to contemporary computing dilemmas involving AI and big data

EAL Student Profile Checklist

Can explain technical concepts using bilingual keyword lists

Can trace algorithm execution with visual support

Can construct extended written responses using sentence frames

Can interpret examination questions using command word glossaries

Can collaborate effectively in mixed-language peer groups

18-Week Curriculum Map

Week	Unit	Topic	Learning Objectives	Assessment
1	Advanced Data Representation	Boolean Algebra — Laws and Theorems	Apply De Morgan's laws, distributive law, absorption law; simplify expressions	Diagnostic test
2	Karnaugh Maps	Construct 3- and 4-variable K-maps; derive minimal SOP/POS expressions	Worksheet	
3	Floating-Point Representation	Represent numbers in IEEE 754 single-precision; convert denary to floating-point	Quiz 1	
4	Normalisation & Errors	Normalise floating-point numbers; calculate absolute/relative error; explain truncation	Worksheet	
5	Advanced Communication	TCP/IP Protocol Stack	Explain layers (application, transport, internet, link); describe encapsulation	Peer presentation
6	Routing Algorithms	Compare distance-vector and link-state routing; trace Dijkstra's algorithm	Quiz 2	
7	DNS Deep Dive	Explain DNS hierarchy, resolution process, record types (A, CNAME, MX)	Worksheet	

Week	Unit	Topic	Learning Objectives	Assessment
8	Email Protocols & Web Technologies	Compare SMTP, POP3, IMAP; explain HTTP methods; basic HTML/CSS structure	Practical task	
9	—	Consolidation Week	Review Weeks 1–8; practice exam questions; formative assessment	Formative test
10	Hardware	Logic Circuits & Boolean Implementation	Design circuits from Boolean expressions; use NAND/NOR universal gates	Practical lab
11	Half-Adders & Full-Adders	Construct truth tables; design ripple-carry adder; explain carry propagation	Quiz 3	
12	Flip-Flops & Sequential Logic	Explain SR, D, JK flip-flops; design simple counters and registers	Worksheet	
13	Microprocessors & Assembly Language	Describe CPU architecture; interpret assembly instructions; trace fetch-decode-execute	Practical lab	
14	System Software	OS Memory Management	Explain paging, segmentation, virtual memory; calculate page table entries	Quiz 4
15	Scheduling Algorithms	Compare FCFS, SJF, RR, priority scheduling; calculate waiting/turnaround times	Worksheet	
16	Interrupts & System Security	Explain interrupt types and handling; describe privilege levels and protection rings	Peer presentation	

Week	Unit	Topic	Learning Objectives	Assessment
17	Artificial Intelligence	Expert Systems & Neural Networks	Describe knowledge base, inference engine; explain perceptron and multi-layer networks	Quiz 5
18	Machine Learning, Big Data & Ethics	Compare supervised/unsupervised learning; explain big data characteristics; evaluate AI ethics	Mock exam	

Assessment Calendar — Semester 1

Week	Assessment	Format	Marks	Weighting
3	Quiz 1: Boolean & Floating-Point	Closed-book	20	10%
6	Quiz 2: Networks & Routing	Closed-book	20	10%
9	Formative Test: Weeks 1 - 8	Closed-book	40	Formative
11	Quiz 3: Hardware & Adders	Closed-book	20	10%
14	Quiz 4: System Software	Closed-book	20	10%
17	Quiz 5: AI & Ethics	Closed-book	20	10%
18	Mock Exam: Paper 1 Style	Closed-book	75	50%

Note: All quizzes include bilingual glossaries. Mock exam follows exact A-Level Paper 1 timing (1 hour 45 minutes) and mark allocation.

Standard Lesson Structure

Each 2-hour lesson follows this structure:

Time	Activity	Purpose
0:00 - 0:10	Starter / Retrieval Practice	Activate prior knowledge; recall key terms from previous lesson

Time	Activity	Purpose
0:10 - 0:35	Direct Instruction	Present new concepts with visual aids, live coding, or circuit simulations
0:35 - 0:45	Worked Example	Teacher models problem-solving; students follow with structured notes
0:45 - 1:15	Guided Practice	Students attempt problems in pairs; teacher circulates with targeted feedback
1:15 - 1:25	Break	—
1:25 - 1:50	Independent Practice	Differentiated tasks based on bronze/silver/gold pathways
1:50 - 1:55	Plenary / Exit Ticket	Quick assessment of learning; preview next lesson

标准课程结构 / Standard Lesson Structure: 每节课包含导入、讲授、示范、练习、独立任务和总结六个环节。Each lesson includes starter, instruction, modelling, practice, independent work, and plenary.

EAL Support Framework

Universal Strategies (All Students)

Visual organisers for all abstract concepts (e.g., TCP/IP stack diagram, memory hierarchy)

Keyword walls with definitions and Chinese translations updated weekly

Sentence starters for all extended-writing tasks

Clear command word definitions on every assessment

Targeted Strategies (EAL Students)

Pre-teaching vocabulary 24 hours before new topics

Bilingual glossaries permitted in all assessments

Extended reading time (+5 minutes per 30-minute assessment)

One-to-one check-ins every fortnight to review understanding

Peer mentoring with bilingual study buddies

Assessment Accommodations

Accommodation	Application
Extra time (+25%)	All timed assessments for EAL students
Bilingual dictionary	All closed-book quizzes and mock exam
Visual prompt sheets	Boolean laws, K-map grouping rules, protocol diagrams

Accommodation	Application
Oral explanation option	Formative assessments and practical demonstrations

Differentiation Framework

Bronze Pathway — Secure Foundation

Target: Students working towards grade C - E

Focus on recall and application of standard procedures

Scaffolded worksheets with partially completed examples

Emphasis on truth tables, straightforward conversions, and definitions

Multiple-choice and short-answer practice

Access to worked solution videos for all homework

Example Task: Complete a truth table for a given 2-input logic circuit. Label the output at each gate.

Silver Pathway — Developing Competence

Target: Students working towards grade B - C

Apply knowledge to unfamiliar problems

Complete K-maps with minimal scaffolding

Explain reasoning in structured paragraphs

Compare and contrast related concepts (e.g., paging vs segmentation)

Debug given code or circuit designs

Example Task: Simplify a Boolean expression using a Karnaugh map, then verify your result using Boolean algebra laws.

Gold Pathway — Advanced Challenge

Target: Students working towards grade A - A*

Synthesise knowledge across multiple topics

Design original solutions to complex problems

Evaluate trade-offs in algorithm and design choices

Construct formal proofs and rigorous arguments

Research and present on cutting-edge developments

Example Task: Design a 4-bit arithmetic logic unit (ALU) using full-adders and logic gates. Calculate propagation delay and suggest optimisation strategies.

Bilingual Glossary — 80+ Terms

English	中文 (Chinese)	Definition
Boolean algebra	布尔代数	A branch of algebra in which variables can have only two values: true (1) or false (0)
Truth table	真值表	A table showing all possible input combinations and their corresponding outputs
De Morgan's laws	德摩根定律	Rules for simplifying Boolean expressions: NOT(A AND B) = (NOT A) OR (NOT B)
Karnaugh map	卡诺图	A visual method for simplifying Boolean expressions by grouping adjacent cells
Sum of products (SOP)	积之和	A Boolean expression formed by ORing multiple AND terms
Product of sums (POS)	和之积	A Boolean expression formed by ANDing multiple OR terms
Floating-point	浮点数	A representation of real numbers using a significand and an exponent
Mantissa	尾数	The fractional part of a floating-point number containing the significant digits
Exponent	指数	The power to which the base is raised in floating-point representation
Normalisation	规范化	Adjusting the mantissa and exponent so the number starts with 1 (binary) or non-zero digit
Truncation error	截断误差	Error caused by discarding less significant digits during conversion or calculation
Absolute error	绝对误差	The difference between the exact value and the approximated value
Relative error	相对误差	Absolute error divided by the exact value, often expressed as a percentage
TCP/IP	传输控制协议/网际协议	A suite of protocols that defines how data is packetised, addressed, transmitted, and received

English	中文 (Chinese)	Definition
Protocol stack	协议栈	A layered architecture where each layer provides services to the layer above
Encapsulation	封装	The process of adding header information as data passes down through protocol layers
Packet	数据包	A unit of data transmitted over a network, consisting of header and payload
Router	路由器	A device that forwards packets between networks based on IP addresses
Routing algorithm	路由算法	A method for determining the optimal path for data to travel across a network
Distance-vector	距离向量	A routing protocol where routers share their entire routing table with neighbours
Link-state	链路状态	A routing protocol where routers share information about their direct connections only
DNS	域名系统	A hierarchical system that translates human-readable domain names into IP addresses
Domain name	域名	A human-readable address used to identify a website (e.g., www.example.com)
IP address	IP地址	A numerical label assigned to each device connected to a computer network
SMTP	简单邮件传输协议	Protocol for sending email messages between servers
POP3	邮局协议第3版	Protocol for retrieving emails from a server; typically deletes after download
IMAP	互联网邮件访问协议	Protocol for accessing emails while keeping them stored on the server
HTTP	超文本传输协议	Protocol for transferring web pages and resources on the World Wide Web

English	中文（Chinese）	Definition
HTML	超文本标记语言	The standard markup language for creating web pages
CSS	层叠样式表	A stylesheet language used for describing the presentation of a document
Logic gate	逻辑门	A physical device implementing a Boolean function (AND, OR, NOT, etc.)
AND gate	与门	Outputs 1 only if all inputs are 1
OR gate	或门	Outputs 1 if at least one input is 1
NOT gate	非门	Outputs the inverse of the input
NAND gate	与非门	Outputs 0 only if all inputs are 1; a universal gate
NOR gate	或非门	Outputs 1 only if all inputs are 0; a universal gate
XOR gate	异或门	Outputs 1 if inputs are different
Half-adder	半加器	A circuit that adds two single binary digits, producing sum and carry
Full-adder	全加器	A circuit that adds three single binary digits (including carry-in), producing sum and carry-out
Ripple-carry adder	行波进位加法器	A multi-bit adder formed by chaining full-adders, where carry propagates sequentially
Flip-flop	触发器	A bistable circuit that stores one bit of data and changes state on a clock pulse
SR flip-flop	SR触发器	Set-Reset flip-flop; basic storage element with set and reset inputs
D flip-flop	D触发器	Data flip-flop; stores the value of D input on clock edge
JK flip-flop	JK触发器	Jack-King flip-flop; versatile with toggle capability when J=K=1

English	中文 (Chinese)	Definition
Register	寄存器	A group of flip-flops used to store multiple bits of data
Counter	计数器	A sequential circuit that cycles through a sequence of states
Microprocessor	微处理器	An integrated circuit containing the CPU functions on a single chip
Assembly language	汇编语言	A low-level programming language with a strong correspondence to machine code
Instruction set	指令集	The set of all commands understood by a particular processor
Fetch-decode-execute	取指-译码-执行	The cycle by which the CPU retrieves, interprets, and carries out instructions
Memory management	内存管理	The process of controlling and coordinating computer memory
Paging	分页	Dividing memory into fixed-size blocks called pages to eliminate external fragmentation
Segmentation	分段	Dividing memory into variable-sized segments based on logical program structure
Virtual memory	虚拟内存	A technique that uses disk space to extend the apparent capacity of RAM
Page fault	页错误	An interrupt raised when a program accesses a page not currently in physical memory
Scheduling	调度	The method by which the OS decides which process runs on the CPU
FCFS	先来先服务	First-Come First-Served scheduling; processes served in arrival order
SJF	最短作业优先	Shortest Job First scheduling; minimises average waiting time
Round Robin	时间片轮转	Scheduling where each process gets a fixed time quantum in cyclic order

English	中文 (Chinese)	Definition
Priority scheduling	优先级调度	Scheduling where the highest priority process runs first
Interrupt	中断	A signal that causes the CPU to pause current execution and handle an event
Interrupt handler	中断处理程序	A routine that processes interrupts and restores normal execution
Privilege level	特权级别	A mechanism for restricting access to sensitive resources (e.g., user vs kernel mode)
Artificial Intelligence	人工智能	The simulation of human intelligence processes by computer systems
Expert system	专家系统	A system that emulates the decision-making ability of a human expert
Knowledge base	知识库	A repository of facts and rules used by an expert system
Inference engine	推理引擎	The component that applies logical rules to the knowledge base to derive conclusions
Neural network	神经网络	A computing system inspired by biological neurons, with interconnected nodes
Perceptron	感知器	The simplest neural network; a single-layer binary classifier
Machine learning	机器学习	A subset of AI where systems learn patterns from data without explicit programming
Supervised learning	监督学习	Training with labelled input-output pairs
Unsupervised learning	无监督学习	Finding patterns in data without predefined labels
Big data	大数据	Datasets characterised by high volume, velocity, and variety
Ethics	伦理	Moral principles governing the development and use of technology

English	中文（Chinese）	Definition
Algorithmic bias	算法偏见	Systematic and repeatable errors in a computer system that create unfair outcomes
Privacy	隐私	The right to control personal information and limit its collection and use
Transparency	透明度	The quality of being open and accountable about how systems operate
Accountability	问责制	The obligation to accept responsibility for the consequences of AI decisions

Pseudocode Reference

Variable Declaration

```
DECLARE variable_name : DATA_TYPE
DECLARE count : INTEGER
DECLARE name : STRING
DECLARE found : BOOLEAN
```

Assignment and I/O

```
variable = expression
INPUT variable
OUTPUT "message", variable
```

Selection

```
IF condition THEN
statements
ELSE IF condition THEN
statements
ELSE
statements
ENDIF
```

Iteration

```
FOR i = start TO end STEP increment
statements
ENDFOR
```

WHILE condition DO

statements

ENDWHILE

REPEAT

statements

UNTIL condition

Arrays and Records

DECLARE array_name : ARRAY[start:end] OF DATA_TYPE

DECLARE student : RECORD

name : STRING

age : INTEGER

grade : CHAR

ENDRECORD

Subroutines

FUNCTION function_name(parameters) RETURNS type

statements

RETURN value

ENDFUNCTION

PROCEDURE procedure_name(parameters)

statements

ENDPROCEDURE

CALL procedure_name(arguments)

result = function_name(arguments)

File Handling

OPENFILE filename FOR READ/WRITE/APPEND

READFILE filename, variable

WRITEFILE filename, data

CLOSEFILE filename

EOF(filename) // returns true if end of file reached

SQL Reference

Basic Queries

SELECT column1, column2 FROM table_name;

SELECT * FROM table_name WHERE condition;

SELECT * FROM table_name ORDER BY column ASC/DESC;

SELECT DISTINCT column FROM table_name;

Aggregation and Grouping

SELECT COUNT(*) FROM table_name;

SELECT AVG(column), MAX(column), MIN(column), SUM(column) FROM table_name;

SELECT column, COUNT(*) FROM table_name GROUP BY column;

SELECT column, COUNT(*) FROM table_name GROUP BY column HAVING condition;

Joins

SELECT * FROM table1 INNER JOIN table2 ON table1.id = table2.id;

SELECT * FROM table1 LEFT JOIN table2 ON table1.id = table2.id;

SELECT * FROM table1 RIGHT JOIN table2 ON table1.id = table2.id;

Data Modification

INSERT INTO table_name (column1, column2) VALUES (value1, value2);

UPDATE table_name SET column1 = value1 WHERE condition;

DELETE FROM table_name WHERE condition;

Boolean Logic Reference

Basic Laws

Law	Expression
Identity	A + 0 = A; A · 1 = A
Null	A + 1 = 1; A · 0 = 0
Idempotent	A + A = A; A · A = A
Inverse	A + A' = 1; A · A' = 0
Commutative	A + B = B + A; A · B = B · A
Associative	(A + B) + C = A + (B + C); (A · B) · C = A · (B · C)
Distributive	A · (B + C) = A·B + A·C; A + (B · C) = (A + B) · (A + C)
Absorption	A + A·B = A; A · (A + B) = A
Double Negation	(A')' = A

De Morgan's Laws

(A + B)' = A' · B'

(A · B)' = A' + B'

"Break the bar, change the sign" — the complement of a sum equals the product of complements, and vice versa.

Gate Symbols and Truth Tables

Gate	Symbol	Truth Table
AND	A —[&]— QB ——/	00→0, 01→0, 10→0, 11→1
OR	A —[⩾1]— QB ——/	00→0, 01→1, 10→1, 11→1
NOT	A —[1]o— Q	0→1, 1→0
NAND	A —[&]o— QB ——/	00→1, 01→1, 10→1, 11→0
NOR	A —[⩾1]o— QB ——/	00→1, 01→0, 10→0, 11→0
XOR	A —[=1]— QB ——/	00→0, 01→1, 10→1, 11→0

Universal Gates

NAND and NOR are universal gates — any Boolean function can be implemented using only NAND gates or only NOR gates.

NOT A = A NAND A

A AND B = (A NAND B) NAND (A NAND B)

A OR B = (A NAND A) NAND (B NAND B)

Resources and Reading List

Essential Textbooks

A-Level Computer Science for AQA — Rouse & O'Byrne (Cambridge University Press)

Computer Science: An Overview — Brookshear & Brylow (Pearson)

Logic and Computer Design Fundamentals — Mano & Kime (Pearson)

Online Resources

Boolean algebra practice: Logisim (http://www.cburch.com/logisim/)

Karnaugh map solver: https://www.charlie-coleman.com/kmaps/

TCP/IP animations: Cisco Networking Academy

Assembly simulator: https://www.peterhigginson.co.uk/AQA/ (AQA Assembler)

Neural network visualiser: TensorFlow Playground

Video Channels

Computerphile (YouTube) — Boolean logic, floating-point, AI concepts

Neso Academy (YouTube) — Digital electronics, OS concepts

3Blue1Brown (YouTube) — Neural networks and linear algebra

Teacher Preparation Checklist

Logisim installed on all lab computers for Weeks 10–13

Assembly simulator accessible for microprocessor lessons

Printed K-map templates available for all students

Bilingual keyword walls updated before each new topic

Past paper questions organised by topic in shared drive

Formative assessment tracker prepared for Weeks 3, 6, 9, 11, 14, 17

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